

Top 50 Advanced Microsoft Excel Functions (with Examples)

1. XLOOKUP

Description: Finds values using flexible lookup

Example: =XLOOKUP(A2, A:A, B:B)

2. INDEX + MATCH

Description: Advanced lookup combination

Example: =INDEX(B:B, MATCH(A2, A:A, 0))

3. SUMIFS

Description: Conditional sum with multiple criteria

Example: =SUMIFS(C:C, A:A, "East", B:B, ">100")

4. COUNTIFS

Description: Conditional count with multiple criteria

Example: =COUNTIFS(A:A, "East", B:B, ">100")

5. AVERAGEIFS

Description: Conditional average

Example: =AVERAGEIFS(C:C, A:A, "East")

6. FILTER

Description: Filters data dynamically

Example: =FILTER(A:C, C:C>100)

7. SORT

Description: Sorts arrays dynamically

Example: =SORT(A2:C10, 3, -1)

8. SORTBY

Description: Sorts based on another column

Example: =SORTBY(A2:C10, C2:C10, -1)

9. UNIQUE

Description: Extracts unique values

Example: =UNIQUE(A2:A100)

10. LET

Description: Assigns variables in formulas

Example: =LET(x, A1+B1, x*2)

11. LAMBDA

Description: Creates custom reusable functions

Example: =LAMBDA(x, y, x*y)(2, 3)

12. TEXTJOIN

Description: Joins text with delimiter

Example: =TEXTJOIN(" ", TRUE, A1:A5)

13. IFS

Description: Multiple logical tests

Example: =IFS(A1>90, "A", A1>75, "B")

14. SWITCH

Description: Evaluates multiple cases

Example: =SWITCH(A1, 1, "Low", 2, "Medium")

15. OFFSET

Description: Returns offset reference

Example: =SUM(OFFSET(A1, 0, 0, 5, 1))

16. INDIRECT

Description: Creates reference from text

Example: =INDIRECT("A"&B1;)

17. AGGREGATE

Description: Advanced aggregation ignoring errors

Example: =AGGREGATE(9, 6, A1:A10)

18. SUBTOTAL

Description: Aggregate visible data only

Example: =SUBTOTAL(109, A1:A100)

19. FREQUENCY

Description: Frequency distribution

Example: =FREQUENCY(A1:A10, B1:B5)

20. MMULT

Description: Matrix multiplication

Example: =MMULT(A1:B2, C1:D2)

21. TRANSPOSE

Description: Switches rows and columns

Example: =TRANSPOSE(A1:C3)

22. SEQUENCE

Description: Generates number sequences

Example: =SEQUENCE(10, 1, 1, 1)

23. RANDARRAY

Description: Generates random arrays

Example: =RANDARRAY(5, 2, 1, 100)

24. PMT

Description: Loan payment calculation

Example: =PMT(5%/12, 60, -10000)

25. FV

Description: Future value calculation

Example: =FV(5%, 10, 0, -1000)

26. NPV

Description: Net present value

Example: =NPV(10%, B1:B5)

27. IRR

Description: Internal rate of return

Example: =IRR(B1:B5)

28. XIRR

Description: IRR for irregular periods

Example: =XIRR(B1:B5, A1:A5)

29. VLOOKUP (advanced)

Description: Lookup with error handling

Example: =IFERROR(VLOOKUP(A1, A:B, 2, FALSE), "")

30. HLOOKUP

Description: Horizontal lookup

Example: =HLOOKUP(A1, A1:D5, 2, FALSE)

31. CHOOSE

Description: Selects from list

Example: =CHOOSE(A1, "Red", "Blue")

32. MATCH

Description: Finds position

Example: =MATCH(A1, A:A, 0)

33. ROW

Description: Returns row number

Example: =ROW(A1)

34. COLUMN

Description: Returns column number

Example: =COLUMN(A1)

35. ISBLANK

Description: Checks empty cells

Example: =ISBLANK(A1)

36. ISERROR

Description: Checks for errors

Example: =ISERROR(A1)

37. IFERROR

Description: Handles errors

Example: =IFERROR(A1/B1, 0)

38. CELL

Description: Returns cell info

Example: =CELL("address", A1)

39. FORMULATEXT

Description: Displays formula

Example: =FORMULATEXT(A1)

40. TEXT

Description: Formats numbers

Example: =TEXT(A1, "\$#, ##0.00")

41. LEFT / RIGHT / MID

Description: Extracts text

Example: =MID(A1, 2, 5)

42. SEARCH

Description: Finds text position

Example: =SEARCH("x", A1)

43. REPLACE

Description: Replaces text

Example: =REPLACE(A1, 1, 3, "New")

44. WEBSERVICE

Description: Fetches web data

Example: =WEBSERVICE("https://example.com")

45. ENCODEURL

Description: Encodes URL strings

Example: =ENCODEURL(A1)

46. MAP

Description: Apply lambda to arrays

Example: =MAP(A1:A5, LAMBDA(x, x*2))

47. REDUCE

Description: Reduce array to single value

Example: =REDUCE(0, A1:A5, LAMBDA(a, b, a+b))